

PROBABILITY, (CONDITIONAL PROBABILITY AND VENN

Name: _____

Date: _____

LO: I can use Venn diagrams to find probabilities including conditional probability.

Conditional probability

Conditional probability is the probability of an event given that another event has already occurred. Venn diagrams help us visualise overlapping groups.

Notation: $P(A|B)$ means the probability of A given B. $P(A|B) = P(A \text{ and } B) \div P(B)$.

Quick examples

●	$P(A B) = P(A \cap B) \div P(B)$	conditional formula
●	A Venn diagram shows overlapping sets	intersection
●	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	addition rule
●	$P(A B) = P(A)$ always	only if independent

Key idea

In a Venn diagram, the overlap shows items in BOTH sets. Conditional probability restricts to a subset.

$$P(A|B) = \frac{n(A \cap B)}{n(B)}$$

Given B, only look at members of B

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – reading a Venn diagram

In a class of 30: 18 like football, 12 like tennis, 5 like both. Find $P(\text{football only})$.

$$\text{Football only} = 18 - 5 = 13$$

$$\text{Total} = 30$$

$$P(\text{football only}) = \frac{13}{30}$$

$$= \frac{13}{30}$$

Subtract the overlap to find the football-only region.

Example 2 – conditional probability

Using the same data: find $P(\text{tennis} | \text{likes football})$.

Given they like football: 18 students

Of these, 5 also like tennis

$$P(\text{tennis} | \text{football}) = \frac{5}{18}$$

$$= \frac{5}{18}$$

Restrict to the football group, then find those who also like tennis.

Example 3 – finding the total

A Venn diagram shows: A only = 8, B only = 12, $A \cap B = 4$, neither = 6. Find $P(A \cap B)$.

$$\text{Total} = 8 + 12 + 4 + 6 = 30$$

$$A \cup B = 8 + 4 + 12 = 24$$

$$P(A \cap B) = \frac{4}{30} = \frac{2}{15}$$

$$= \frac{2}{15}$$

A union B includes everything inside at least one circle.

Example 4 – completing a Venn diagram

60 students: 35 study French, 25 study Spanish, 10 study both. How many study neither?

French only = $35 - 10 = 25$, Spanish only = $25 - 10 = 15$

$$\text{Total in circles} = 25 + 10 + 15 = 50$$

$$\text{Neither} = 60 - 50 = 10$$

$$= 10 \text{ students}$$

Add all regions and subtract from total.

YOUR TURN – PRACTICE (deliberate practice)

1. Reading Venn diagrams

A class of 40: $A = 22$, $B = 18$, $A \cap B = 8$, neither = 8.
Find:

- a) $P(A \text{ only})$
- b) $P(B \text{ only})$
- c) $P(A \cap B)$
- d) $P(\text{neither})$

2. Conditional probability

Using: $A = 22$, $B = 18$, $A \cap B = 8$, total = 40.

- a) $P(B|A)$
- b) $P(A|B)$
- c) $P(\text{not } B|A)$
- d) $P(A|\text{not } B)$

3. Completing Venn diagrams

Find the missing values.

- a) Total 50, $A = 30$, $B = 20$, $A \cap B = 7$. Neither?
- b) Total 80, $A = 45$, $B = 35$, neither = 15.
 $A \cap B$?
- c) Total 100, $A \text{ only} = 25$, $B \text{ only} = 30$,
neither = 20. $A \cap B$?
- d) Total 60, $A = 40$, $A \cap B = 12$, neither = 5. B ?

4. Word problems

Draw a Venn diagram and find the probability.

- a) 50 people: 30 drink tea, 25 drink coffee, 15 drink both. $P(\text{tea only})$?
- b) Same data: $P(\text{coffee} | \text{drinks tea})$?
- c) Same data: $P(\text{neither})$?
- d) Same data: $P(\text{tea} | \text{drinks coffee})$?

5. Mixed conditional probability

Use the given information.

a) $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cap B) = 0.2$

d) Are A and B independent?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $P(A|B)$ means restrict to set B then find $P(A)$ ☐

Correct answer: _____

Reason: _____

b) $P(A \cap B) = P(A) + P(B)$ always ☐

Correct answer: _____

Reason: _____

c) If $P(A|B) = P(A)$ then A and B are independent ☐

Correct answer: _____

Reason: _____

d) $P(A|B)$ is the same as $P(B|A)$ ☐

Correct answer: _____

Reason: _____

e) $P(A \cap B) = P(A) + P(B) - P(A \cap B)$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) In a school of **200** students: **120** study Maths, **90** study Science, **50** study both. Draw a Venn diagram. Find $P(\text{Maths} | \text{Science})$ and $P(\text{Science} | \text{Maths})$. Are Maths and Science independent events?

Expression: _____

Simplified: _____

b) A medical test is **95% accurate**. A disease affects **1%** of people. If a person tests positive, what is the probability they actually have the disease? (Hint: consider 10,000 people.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

